

NUMERICAL PROBLEMS IN ELECTROSTATIC POTENTIAL

MANDATORY PROBLEMS

1. The electric field at a point due to point charge is 30 N/C & electric potential at the same point is 15 J/C. Calculate the distance of the point from the charge and the magnitude of the charge.
2. Three point charges $Q_1 = 25 \mu\text{C}$, $Q_2 = 50 \mu\text{C}$ and $Q_3 = 100 \mu\text{C}$, are kept at the corners A, B and C respectively of an equilateral triangle ABC having each side equal to 7.5 m. Calculate the total electrostatic potential energy of the system.
3. ABCD is a square of side 0.2 m. Charges of $2 \times 10^{-9} \text{ C}$, $4 \times 10^{-9} \text{ C}$ and $8 \times 10^{-9} \text{ C}$ are placed at the corners A, B and C, respectively. Calculate the work to be done to transfer a charge of $2 \times 10^{-9} \text{ C}$ from the corner D to the centre of the square.
4. How much work is done in taking an electron around a nucleus in a circular path?
5. In an electric dipole, what is the locus of a point of zero potential?
6. An electron revolves around the nucleus of hydrogen atom in a circular orbit of radius $5 \times 10^{-11} \text{ m}$. Calculate: (i) intensity of electric field of the nucleus at the position of the electron (ii) electrostatic potential energy of the hydrogen nucleus and electron system.
7. How much kinetic energy is gained by an electron, initially at rest, when it moves through a potential difference of 1000 V? Leave your answer in joule.
8. What will be the nature of the electric potential at a place where the electric field $E = 0$?
9. Two point charges of 10 C each are kept at distance of 3m in vacuum. Calculate their electrostatic potential energy.
10. Three equal charges of $5.0 \mu\text{C}$ each are placed at the three vertices of an equilateral triangle of side 5.0 cm each. Calculate the electrostatic potential energy of the system of charges.

PRACTICE PROBLEMS

1. Charges of 10 nC, - 20 nC, 30 nC and 20 nC are placed at the corners A, B, C and D of a square ABCD of side 1 m. Find the potential at the intersection of the diagonals.
2. Calculate the work to be done to displace an electron which is at a distance of 5 cm from a charge of $20 \mu\text{C}$ to a distance 10 cm.